

Assignment 3

Team Name: Panem

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Instructions

- Please **work in your teams**.
- Please go to Assignment 3 on the Assignments page in Canvas and use the upload link to submit your answers in a PDF file named **TEAMNAME_A3.pdf**. Please also submit your spreadsheet from Part 1 as **TEAMNAME_A3.xlsx**.
- To receive full credit, please show your calculations for quantitative questions and explain your reasoning for qualitative questions.
- Refer to the following resources in completing the assignment: *Gleacher Game Case Brief*, *Gleacher Game Player Manual*, *Market Research for Practice Game*, *Gleacher Game Product Design Guide*, and *Production Model Template*.

Part 1: Analyze the Bench, Production Line, and Automated Cell Factory Types

These questions are related to the Practice Game.

1. Using the *Production Model Template* posted on Canvas, fill in the formulas in the tables to calculate daily throughput, manufacturing overhead per unit, and production cost per unit for each type of factory. Production cost is materials cost + manufacturing overhead. Capital investment (K) is captured as the cost of the factory and labor (L) is captured as recurring expenses in this game. You can refer to the *Gleacher Game Case Brief* for more information on the Cobb Douglas Production Function and how it is applied in the simulation.
 - a) Submit your spreadsheet with this assignment.
 - b) Use your spreadsheet to analyze the effects of changing labor, capital, and batch size on throughput and manufacturing overhead per unit for the three types of factories. Change one factor at a time. State three valuable insights that you have gained from your analysis.

Baseline Comparison (K = minimum, L = \$2,500/day, batch = 100)

Factory	K	Daily Throughput	OH/unit	Production Cost/unit
Bench	\$100,000	9.04 units	\$278.45	\$378.45
Production Line	\$500,000	34.36 units	\$75.43	\$175.43
Automated Cell	\$3,000,000	83.66 units	\$36.45	\$136.45

Updated Comparison (K = \$3,000,000, L = \$2,500/day, batch = 100)

Factory	K	Daily Throughput	OH/unit	Production Cost/unit
Bench	\$3,000,000	12.68 units	\$240.41	\$240.41
Production Line	\$3,000,000	52.40 units	\$58.19	\$158.19
Automated Cell	\$3,000,000	83.66 units	\$36.45	\$136.45

Three Valuable Insights from Sensitivity Analysis

- Insight 1: Capital vs. labor sensitivity differs dramatically by technology. The exponent ratio α/β shows which input drives output. Bench ($0.10 / 0.85 = 0.12$) is labor-driven.

Automated Cell ($0.80 / 0.30 = 2.67$) is capital-driven. Each production system responds far more strongly to either capital or labor, so spending should prioritize the input that drives greater output gains.

- Insight 2: Setup time crushes small-batch efficiency, especially for Automated Cell. Set up time for Bench is only 0.05 days, vs. for Cell it's 1 day. Longer setup cycles drastically lower effective throughput, and larger batches help absorb setup costs to improve overall performance.
 - Insight 3: Economies of scale favor advanced technologies but require demand to justify them. According to the table above, daily throughput is much higher and per unit cost is much lower for "Automated Cell", but minimum capital requirement is \$3,000,000. More automated systems yield stronger scale benefits but need higher upfront capital, which only makes financial sense with sufficient volume.
2. Lay out your metrics and decision rules for determining which factory type to use under which circumstances. Discuss how your preferred factory type might change with the evolution of your business and competitive environment. Also discuss how you will decide how much capital and labor to use in each factory.

Decision Rules for Factory Type Selection

Scenario	Recommended Factory	Rationale
Demand < 10 units/day, early game uncertainty	Bench	Low capex, flexible, cheap exit
Demand 10–50 units/day, validated product	Production Line	Best overhead/unit at moderate scale
Demand > 50 units/day, confident in 4-year horizon	Automated Cell	Dominant unit economics at scale
Unknown demand, short horizon	Bench	Preserve optionality

Evolution of Preferred Factory Over Game

1. Early game (Days 1–200): Bench — uncertainty about product-market fit, Bass curve still in innovator phase. Preserve capital.
2. Mid game (Days 200–1000): Transition to Production Line once product is validated and demand >15 units/day. Balanced K and L requirements.
3. Late game (Days 1000+): Automated Cell if pursuing a high-volume market with remaining horizon justifying payback.

4. Competitive response: If competitors deploy Automated Cell early to drive volume, our counter is differentiation (unique feature combinations) or focus (premium markets where WTP supports higher prices regardless of production cost).

How to Decide Capital and Labor Levels

Bench ($\alpha=0.10$, $\beta=0.85$): Labor-oriented strategy. Hold capital steady and expand labor to match supply with demand until demand and supply reach an equilibrium. Avoid excessive capital investment, as capital contributes little to additional throughput.

Production Line ($\alpha=0.30$, $\beta=0.75$): Balanced capital and labor approach. Begin with a moderate capital base and adjust labor to match demand. Increase capital only when labor is fully utilized and throughput stops rising.

Automated Cell ($\alpha=0.80$, $\beta=0.30$): Capital-oriented strategy. Prioritize capital investments to drive throughput gains, maintaining moderate labor levels. Capital delivers the strongest marginal returns for this system.

Part 2: Analyze Heart View Contribution Margin

These questions are related to the Practice Game.

Starting parameters:

- Capital invested in the Bench factory = \$100,000
 - Labor expenditure at the Bench factory = \$2,500 per day
 - Materials cost = \$100 per unit
 - Batch size for production and shipping = 100 units
 - Shipping cost = \$5,000 per container
 - Retail price = \$700 per unit
3. Analyze contribution margin per unit and per day in each of the following scenarios. Assume that all units manufactured are sold.
 - a) Assume the starting parameters in the practice game, as laid out above.

	Per Unit	Per Day
Revenue	700	6,330.76

Manufacturing Overhead	278.45	2,518.32
Materials	100	904.39
Shipping	20	180.88
Handling (\$10 per unit)	10	90.44
Commission (20% of revenue)	140	1,266.15
Contribution Margin before Tax	151.55	1,370.58
Tax (35% tax rate)	53.04	479.70
Contribution Margin after Tax	98.51	890.88

- b) Assume the starting parameters in the practice game, as laid out above, except increase the price by 20%. Comment on the results of this scenario compared to those in (a).

	Per Unit	Per Day
Revenue	840	7596.93
Manufacturing Overhead	278.45	2518.32
Materials	100	904.39
Shipping	20	180.88
Handling (\$10 per unit)	10	90.44
Commission (20% of revenue)	168	1519.38
Contribution Margin before Tax	263.55	2383.50
Tax (35% tax rate)	92.24	834.23
Contribution Margin after Tax	171.31	1549.28

- c) Assume the starting parameters in the practice game, as laid out above, except decrease the price by 20%. Comment on the results of this scenario compared to those in (a).

	Per Unit	Per Day
Revenue	560	5064.61
Manufacturing Overhead	278.45	2518.32
Materials	100	904.39
Shipping	20	180.88
Handling (\$10 per unit)	10	90.44
Commission (20% of revenue)	112	1012.92
Contribution Margin before Tax	39.55	357.66
Tax (35% tax rate)	13.84	125.18
Contribution Margin after Tax	25.71	232.48

- d) Assume the starting parameters in the practice game, as laid out above, except change the factory's daily labor expenditure to \$4,000. Comment on the results of this scenario compared to those in (a).

	Per Unit	Per Day
Revenue	700	9418.78
Manufacturing Overhead	298.64	4018.32
Materials	100	1345.54
Shipping	20	269.11
Handling (\$10 per unit)	10	134.55
Commission (20% of revenue)	140	1883.76
Contribution Margin before Tax	131.36	1767.51
Tax (35% tax rate)	45.98	618.63
Contribution Margin after Tax	85.38	1148.88

- e) Assume the starting parameters in the practice game, as laid out above, except change the factory's daily labor expenditure to \$3,000. Comment on the results of this scenario compared to those in (a).

	Per Unit	Per Day
Revenue	700	7386.37
Manufacturing Overhead	286.04	3018.32
Materials	100	1055.20
Shipping	20	211.04
Handling (\$10 per unit)	10	105.52
Commission (20% of revenue)	140	1477.27
Contribution Margin before Tax	143.96	1519.03
Tax (35% tax rate)	50.38	531.66
Contribution Margin after Tax	93.57	987.37

Part 3: Design a New Product for the MD Fertility (Estrogen) Market

These questions are related to the Practice Game.

4. Using the *Market Research for Practice Game* and the *Gleacher Game Product Design Guide* documents, design a product that would appeal to the MD Fertility (Estrogen) market.
 - a) How much would it cost to develop this product from the ground up? How many days would it take to develop? What would be the materials cost of production? Please specify the product features and your calculations.

Market Context

- Target market: MD Fertility (Estrogen)
- Core feature required: Hormone (Estrogen)
- Willingness to pay: \$575 – \$965 per unit
- Market size: 10,000 – 20,000 customers per region
- Bass parameters: $p = 0.0002$, $q = 0.0035$, DSO = 30 days
- Deal breakers: None. Slight preference for wrist-worn devices.

Product Design: "Panem E"

To maximize margin, we design a minimal-feature Estrogen monitor. Only the required Hormone (Estrogen) feature is selected; all other attributes are set to "None".

Attribute	Selection	Days	Design Cost	Materials/unit
Heartbeat	None	3	\$1,000	\$0
Blood vessel	None	3	\$1,000	\$0
Dissolved gasses	None	3	\$1,000	\$0
Toxicology	None	3	\$1,000	\$0
Hormone	Estrogen	60	\$75,000	\$60
Metabolic	None	3	\$1,000	\$0

"None" has \$0 cost for a brand-new product (ground up)

- Development time: 60 days
- Development cost: \$75,000
- only for adding Estrogen
- Materials cost: \$60 per unit
- total development time is 60 days.

- b) In the game, each product is developed as a derivative of an existing product. You only incur the development cost and time for changes in component features that are

inherited from the parent product. Any features that stay the same do not require redevelopment. The new product's materials cost is based on the materials costs of all its component features.

How much would it cost you to develop this product as a derivative of Heart View? How many days would it take to develop? What would be the materials cost of production? Please specify the product features and your calculations.

Assumption: Heart View, based on the Case Brief description of a "wearable heart monitor that performs sophisticated, continuous diagnosis," is assumed to have Heartbeat = Temporal and all other attributes set to None. This should be checked in the game by clicking Heart View under HQ → Products.

Changes from Heart View to Panem E

- Heartbeat: Temporal → None
- Development cost: \$1,000
- Development time: 3 days
- Materials cost savings: \$25 per unit
- Hormone: None → Estrogen
- Development cost: \$75,000
- Development time: 60 days
- Additional materials cost: \$60 per unit
- All other attributes: unchanged
- None → None

Derivative design summary

- Development time: 60 days
- This is the longer of the two changes: 3 days for removing heartbeat and 60 days for adding Estrogen.
- Development cost: \$76,000
- This includes \$1,000 for removing heartbeat and \$75,000 for adding Estrogen.
- Materials cost: \$60 per unit
- Removing Temporal heartbeat saves \$25 per unit, while adding Estrogen adds \$60 per unit.
- So the net change is: $0 - 25 + 60 = 35$

If Heart View originally had only Temporal heartbeat with \$25 materials cost, then the new product's materials cost becomes \$60 per unit.

- c) Fill in the following table for the new product on a per unit basis. Assume the starting factory parameters in the practice game, as laid out above. Assume container shipping to your DC. Assume a reasonable price based on the product's variable cost and the

range of willingness to pay for customers in this market that is provided in *Market Research for Practice Game*.

	Per Unit
Revenue	\$731.00
Manufacturing Overhead	\$278.45
Materials	\$60.00
Shipping / mail	\$20.00
Handling (\$10 per unit)	\$10.00
Commission (20% of revenue)	\$146.20
Contribution Margin before Tax	\$216.35
Tax (35% tax rate)	\$75.72
Contribution Margin after Tax	\$140.63

Pricing decision: Assume that WTP is evenly distributed between \$575 and \$965. The expected profit from each arriving customer is:

$$P(\text{buy}) \times \text{CM}_{\text{per_sale}} = [(965 - P) / 390] \times (0.8P - \$278.45 - \$60 - \$20 - \$10)$$

After taking the derivative and setting it equal to zero:

$$\text{Optimal } P^* = (\$398.45 + 772) / 1.6 = \$731.53$$

So, the optimal price is about \$731.53.

At $P = \$731$, the probability of purchase is 60%.

Recommended price: \$731 per unit.

Parameters: Bench factory with $K = \$100K$, $L = \$2,500$ per day, batch size = 100, in-region container shipping = \$50 per unit, and materials cost = \$60 per unit.

For the MD Fertility (Estrogen) market, we designed a new product with Hormone = Estrogen as the main feature, while all other features are set to None. If the product is developed from scratch, the development cost is \$75,000. The development time would be 60 days, and the materials cost would be \$60 per unit. If this product is developed as a derivative of Heart View, and if we assume that Heart View has Temporal heartbeat and all other features set to None, then the new product would require removing the heartbeat feature and adding Estrogen. In this case, the development cost would be \$76,000, the development time would still be 60 days, and the materials cost would remain \$60 per unit. Using the starting Bench factory parameters, a batch size of 100, and in-region mail shipping, if the selling price is set at \$731, the contribution margin before tax would be \$216.35 per unit, and the contribution margin after tax would be

\$140.63 per unit. Based on the assumption that willingness to pay is uniformly distributed between \$575 and \$965, the optimal price is about \$731 per unit.